

Lecture 4: Biodegradation Process

What is biodegradation?

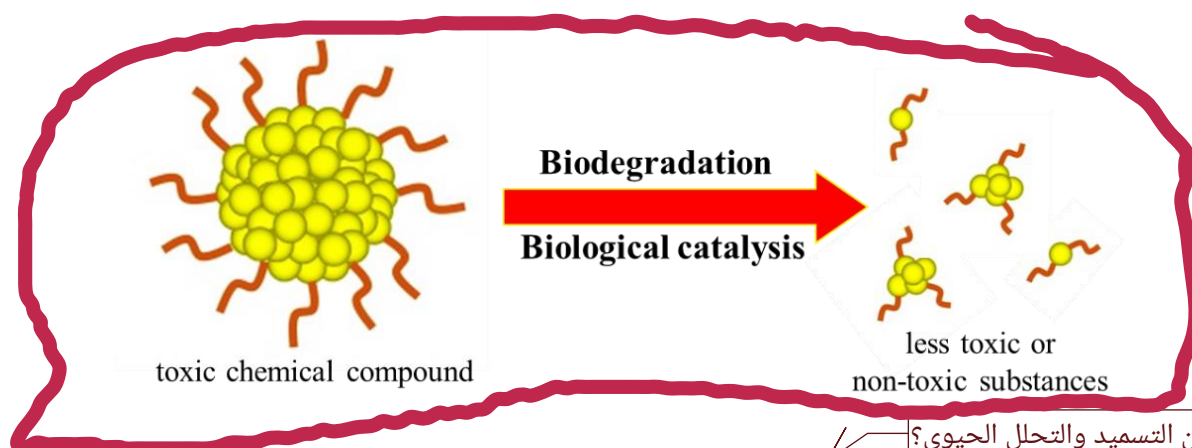
ما هو التحلل البيولوجي؟

Biodegradation is a biologically catalyzed process by which exogenous organic substances are broken down into smaller compounds by **living organisms**. Biodegradation is **nature's way** of recycling wastes, or breaking down organic matter into nutrients that can be used and reused by other organisms which supplement the environment with nutrients.

The term **biodegradation** is strongly associated with bioremediation because biodegradation is the mechanism, while bioremediation is the application of that mechanism for environmental cleanup.

Breakdown of organic compounds could involve two or more kinds of organisms acting in **cooperation**. Such degradative abilities are employed to:

- ▶ convert waste materials into more acceptable and manageable forms or
- ▶ to produce useful end products such as production of biofuel and produce acetic acid during intermediate fermentation stages.



ما هو الفرق بين التسميد والتحلل الحيوي؟

What is the difference between composting and biodegradation?

- Composting is a natural breaking down process of organic wastes by microorganisms under **controlled conditions**. For example, compost the disposal for plant wastes or domestic green waste such as vegetable peelings. A compostable product needs to biodegrade within 180 days in a composting system and leave no toxicity in the soil.
- Biodegradation is a **natural** breaking down process of organic waste by microorganisms. For example, breakdown of vegetable products. Biodegradation may take many years and depends on where the product is disposed of.

Time required for biodegradation of different materials

The time required for biodegradation of different materials is different. Here is the time that it takes for some examples:

صح وخطا او اختيارات

Product	Time
Vegetables	5 days – 1 month
Paper	2 – 5 months
Cotton T-shirt, Orange peels	6 months
Tree leaves	1 year
Wool socks	1 – 5 years
Plastic-coated paper milk cartons	5 years
Leather shoes	25 – 40 years
Tin cans	50 – 100 years
Aluminum cans	80 – 100 years
Glass bottles	1 million years
Plastic bags	500 years to forever

Biodegradable and non-biodegradable pollutants are differing in their biodegradability:

الملوثات القابلة للتحلل البيولوجي وغير القابلة للتحلل بينهم

Biodegradable Pollutants

can be broken into simple and harmless substances by the activity of microorganisms.

Examples:

paper, wood, clothes, cattle dung, sewage, agricultural residue, leather, etc.



Non-Biodegradable Pollutants

Products which are not naturally biodegraded by the ecosystem of our planet are those which cause a long-term risk.

Examples:

Computer components, water bottles, plastic bags, metals, tin cans, metal products, rubber products, man-made fiber products including some clothing, chemical products including detergents.



Type of biodegradation

نوع التحلل البيولوجي تعاريف مهمه مهمه

(1) Biomineralisation

Mineralisation is the process in which microorganisms feed on organic compounds and by a chemical process, reduce them to inorganic material such as water, carbon dioxide, and other inorganic compounds. In mineralisation process **total degradation** of the organic matter occurs. For example, certain hydrocarbon-degrading bacteria oxidize Hexane (C_6H_{14}) from crude oil fully: $2C_6H_{14} + 19O_2 \rightarrow 12CO_2 + 14H_2O$

(2) Biotransformation

In biotransformation process, organic matter is **not degraded totally**. Some part of it is degraded and another part is converted into other smaller chain organic compounds. The converted smaller chain organic compounds may be either **toxic or non-toxic**. In the case of the pesticide DDT, the biotransformation yields **more toxic compound**.

كيف يحدث التحلل البيولوجي؟

How does biodegradation occur?

Understanding the process of biodegradation requires an understanding of the microorganisms that make the process work.

The microorganisms transform the substance through **metabolic or enzymatic processes**.

Two types of enzymes from the microorganisms are involved in biodegradation process:

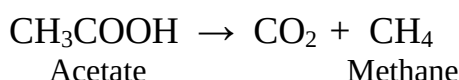
- 1) Extracellular depolymerases
- 2) Intracellular depolymerases

هناك نوعان من الإنزيمات من الكائنات الحية الدقيقة متورطان في عملية التحلل

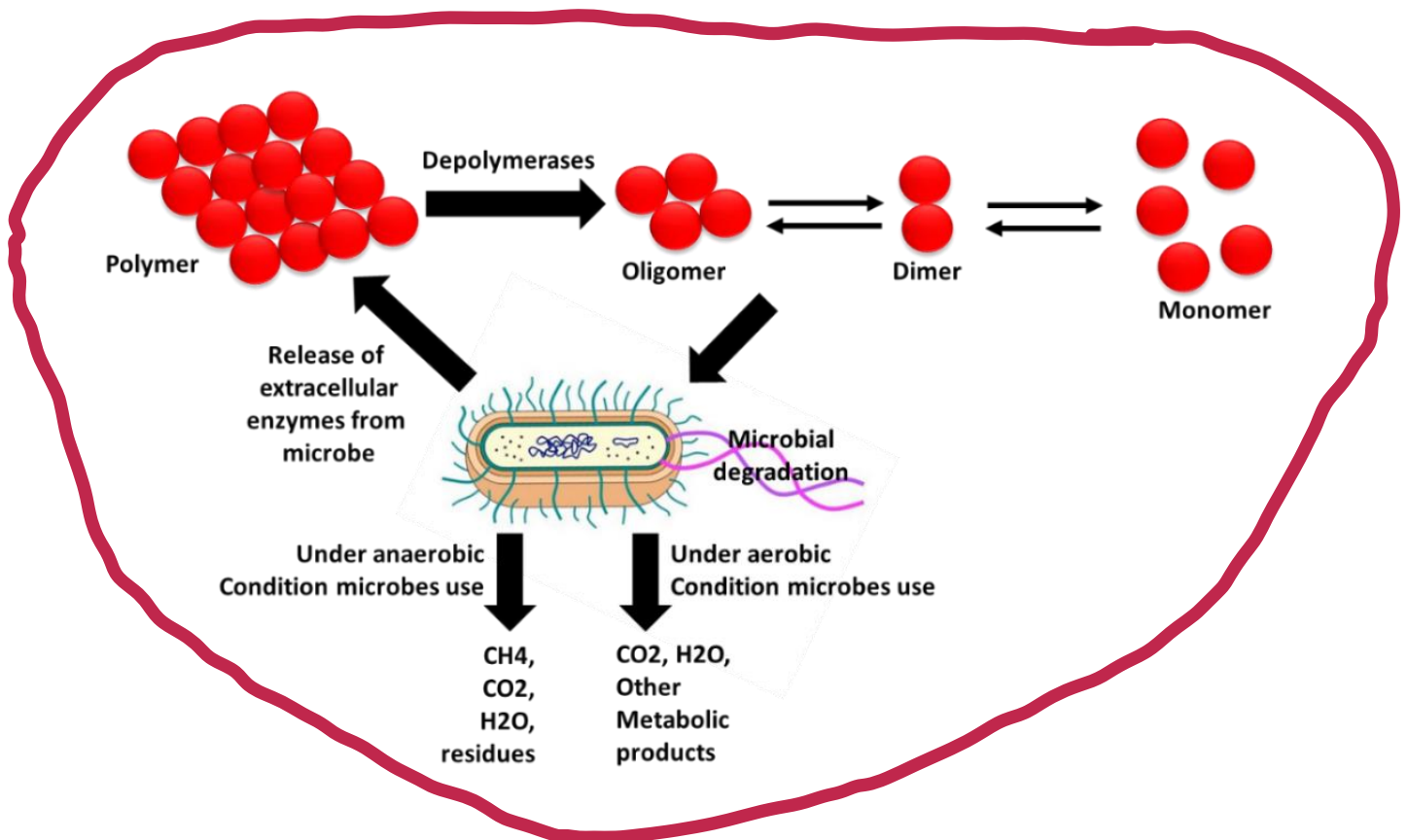
Extracellular enzymes first breakdown the complex polymers giving short chains that are small enough to permeate through the cell walls to be utilized as carbon and energy sources. This process is called **depolymerization**.

إزالة البلمرة.

When the end products are carbon dioxide, water (under aerobic conditions) or methane (under anaerobic conditions), the process is called **mineralization**. Different end products are formed depending upon the degradation pathway. For example, Methanogenesis process occurs in landfills:



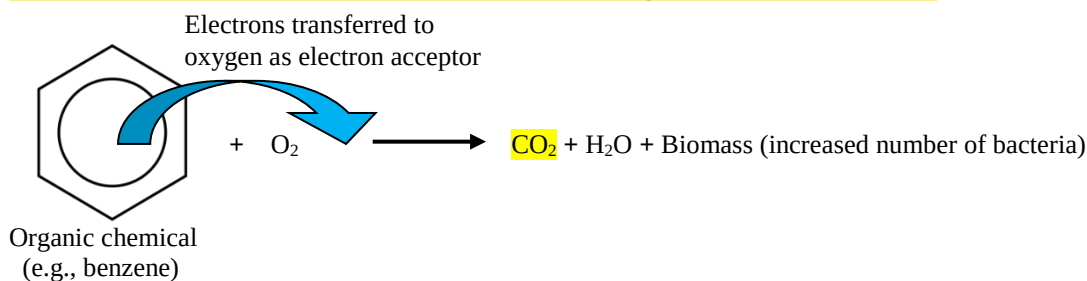
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الأيض الهوائي

Aerobic metabolism

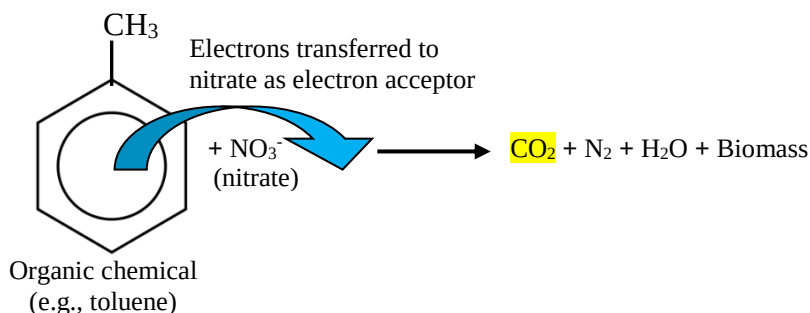
Microbes use O₂ in their metabolism to degrade contaminates:



الأيض اللاهوائي

Anaerobic metabolism

Microbes substitute another chemical for O₂ to degrade contaminates (e.g., nitrate, iron):

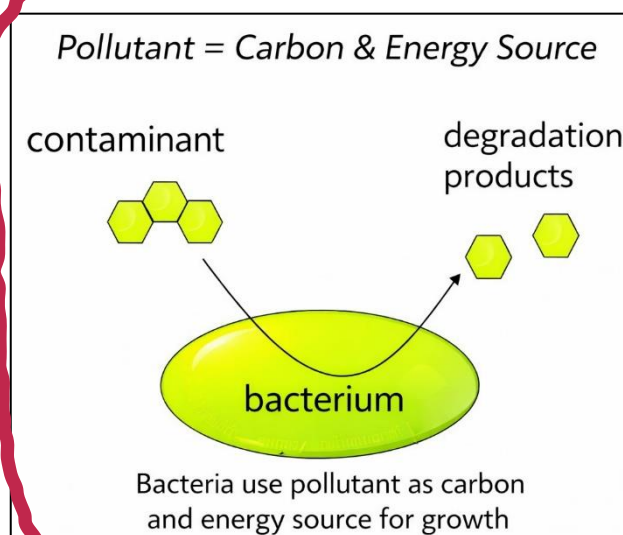
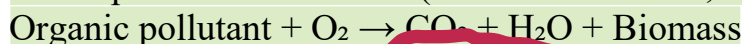


Biodegradation is based on two processes: **Growth-Linked Biodegradation and cometabolism:**

يعتمد التحلل البيولوجي على عمليتين: التحلل البيولوجي المرتبط بالنمو واستقلاب المذنب:

(1) Growth-Linked Biodegradation (Pollutant = Carbon & Energy Source)

In growth-linked biodegradation, the pollutant serves as the primary carbon and energy source. Microorganisms grow while degrading the compound, often resulting in complete mineralization (conversion to CO₂, H₂O, and biomass).



(1) التحلل البيولوجي المرتبط بالنمو

أمثلة

Examples:

(a) Crude Oil degradation

Crude oil hydrocarbons such as alkanes and simple aromatics are commonly degraded through growth-linked metabolism. Specialized marine and soil bacteria utilize these hydrocarbons as sole carbon sources. As degradation proceeds, microbial biomass increases and hydrocarbons are mineralized to CO₂ and H₂O.

(b) Atrazine

Is a xenobiotic compound which was widely used as herbicides for the control of broad leaf and grassy weeds. Atrazine is degraded by *Pseudomonas* sp. isolated from a herbicide spill site in Minnesota/USA through growth-linked metabolism. *Pseudomonas* sp. metabolizes atrazine to carbon dioxide and ammonia and uses atrazine as the sole nitrogen source for growth.

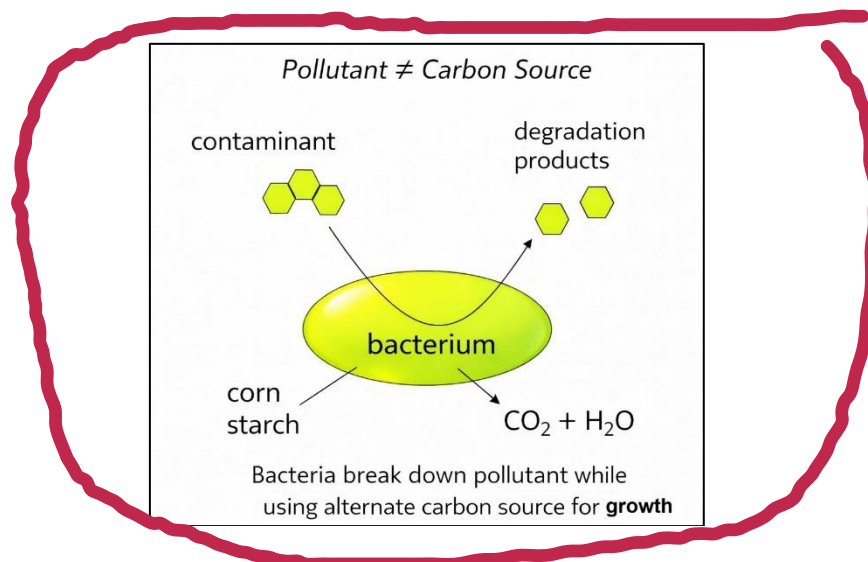
(c) Triclosan

Is an antimicrobial agent used in a wide range of consumer products, such as soaps, toothpastes, hand creams. *Pseudomonas* strains are typically resistant to triclosan. A researcher from Cornell University/USA screened a triclosan-degrading microorganisms by using samples from a municipal wastewater treatment plant. A stable bacterial community consisting of six microorganisms was capable of using triclosan as the sole source of carbon and energy, removing over 90% of the supplied triclosan with 35% mineralization of the compound.

(2) Cometabolism (Pollutant $\neq$ Carbon Source)

مصدر الكربون $\neq$ الملوث (2) cometabolism

In cometabolism, the pollutant is not used as a carbon or energy source. Instead, microorganisms grow on a separate primary substrate (e.g., glucose, methane, or natural organic matter). Enzymes produced for metabolism of the primary substrate incidentally transform the pollutant. Degradation is often partial.



Examples

(a) Oil (Complex Hydrocarbons)

While simple hydrocarbons are degraded via growth-linked metabolism, complex polycyclic aromatic hydrocarbons (PAHs) may undergo cometabolic transformation. Microorganisms growing on simpler hydrocarbons produce oxidative enzymes that incidentally modify complex compounds, often resulting in partial degradation.

(b) Pesticides

Persistent pesticides such as DDD (Dichloro diphenyl dichloroethane) cannot support microbial growth. Microorganisms grow on natural soil organic matter and produce enzymes that accidentally transform DDT into less chlorinated intermediates. Complete mineralization is rare. This is a classical cometabolic process.

(c) Plastics

Most plastics such as polyethylene are not primary carbon sources for microorganisms. In soil environments, microbes grow on organic matter and produce oxidative enzymes (e.g., peroxidases, monooxygenases) that slowly oxidize plastic polymers. This results in partial degradation but does not support microbial growth.

Requirements for biodegradation

There are six requirements for biodegradation should be available:

- 1) The presence of the appropriate microorganisms,
- 2) An energy source for cell maintenance and growth.
- 3) A carbon source.
- 4) An electron acceptor such as O_2 , NO_3^- , SO_4^{2-} , etc.
- 5) Nutrients as nitrogen, phosphorus, calcium, magnesium, iron, trace elements.
- 6) Appropriate environmental conditions e.g., temperature, pH, etc.

متطلبات التحلل البيولوجي